Vetoed — an autonomous options agent where the AI is the least-trusted component

Alpaca AI Trading Agents Hackathon · paper trading only
chong1120.github.io/Vetoed · github.com/Chong1120/Vetoed

The problem

If a language model can pick the trade, choose the size and send the order, one hallucinated contract is a real loss — and you cannot unit-test a language model. You *can* unit-test the code that decides whether to obey it. So in Vetoed the model's entire authority is this: **pick one item from a list it did not write.**

What it trades

Defined-risk credit spreads on SPY, QQQ, IWM and AAPL. Two legs always, sent as one atomic multi-leg order, so the maximum loss is a known number before the order exists and a naked short is structurally impossible.

How the agent works — the workflow

1. **Screen.** A deterministic screener builds every valid spread from the live chain and prices each one twice through the same model, changing only the volatility input: 20-day realised against the market's implied. The gap is the premium being harvested, in dollars. Anything under $2.00 is discarded. This is our own model-derived measure, deliberately not the academic variance risk premium, which is defined and measured differently.
2. **Select.** The LLM picks one candidate from that shortlist. Its answer is validated against the list, so an invented contract is unexecutable, and the position size it suggests is discarded.
3. **Gate.** A deterministic risk module sizes the trade and can overrule the model at any confidence. Eight hard gates: 5% of equity per position, 25% across all, a 3% daily-loss halt, a 25-contract cap, plus liquidity, DTE, volatility and structural checks. Any one rejects the trade outright.
4. **Execute.** Orders go through Alpaca's own MCP server — the single write path in the system — as one atomic multi-leg order carrying a deterministic SHA-1 order ID, so a restart cannot open the same spread twice.
5. **Reconcile.** Every cycle re-reads positions and open orders from Alpaca, never from local state. The broker is the source of truth.

How it uses Alpaca

Alpaca MCP server for all order placement; Alpaca options chain, Greeks and quotes for screening; Alpaca positions and orders for reconciliation; Alpaca's market clock to decide whether to trade at all. The process refuses to start unless ALPACA_PAPER_TRADE=true.

What the dashboard shows that most do not

Every rejection, by the reason that caused it. A typical cycle measures the chains and declines around 1,500 spreads — open interest below the liquidity floor, premium under ten cents, delta outside the band, edge under $2.00 — each reason counted and naming real contracts. Each position also carries a note the model writes from that position's journal entry, using figures quoted from the record; it is never allowed to calculate.

Operating record

Unattended on GitHub Actions, started by a Vercel cron: **148 cycles, 749 candidates screened, 12 orders sent — 1.6% of what it looked at.** Currently 5 open spreads and 7 closed, −$406 realised on a $100,000 paper account. **242 tests**, most of them on the gates.

What I am not claiming

Quotes are indicative, not exchange best bid and offer. Skew is not modelled. Twenty-day realised volatility is an estimator, not a forecast. Four correlated tickers is not real diversification. The exit thresholds are unvalidated. A contest week proves nothing about profitability — the research motivates the idea; it does not validate this system.